hyper2kvm KubeVirt Demo

VMware Photon OS VMDK to K3s + KubeVirt — End-to-End

Live demonstration: VMDK conversion with offline fixes, dual deploy to libvirt domain XML AND KubeVirt VirtualMachine on K3s, PVC upload via CDI, VM running on Kubernetes with VirtIO disk and masquerade networking.

Photon OS 5.0 | K3s v1.34 | KubeVirt v1.7.2 | CDI v1.64 | VirtIO | PVC Upload | VM Running

Step 1-2: Cluster & Configuration

STEP 1 — K3s + KubeVirt Cluster

```
$ kubectl get nodes -o wide
NAME                   STATUS   ROLES           VERSION        OS-IMAGE     KERNEL
ssahani-thinkpad...    Ready    control-plane   v1.34.5+k3s1   Fedora 43    6.19.8-200.fc43

$ kubectl get pods -n kubevirt
NAME                               READY   STATUS    AGE
virt-api-9c497cb7d-tg7sj           1/1     Running   3h
virt-controller-6f4f5ff947-vfkbs   1/1     Running   3h
virt-controller-6f4f5ff947-vgfh5   1/1     Running   3h
virt-handler-wljr2                 1/1     Running   3h
virt-operator-d9474c954-27n9d      1/1     Running   3h
virt-operator-d9474c954-phzds      1/1     Running   3h

$ kubectl get pods -n cdi
cdi-apiserver-6cc6bcdc46-gdm78     1/1     Running   26m
cdi-deployment-59945cb874-pltzd    1/1     Running   26m
cdi-operator-697fd6d5b4-89t28      1/1     Running   26m
cdi-uploadproxy-7787b9cc65-kbxt7   1/1     Running   26m
```

STEP 2 — Migration YAML Config

```
cmd: local
vmdk: ./photon.vmdk
output_dir: ./out-kubevirt-demo
flatten: true
out_format: qcow2
fstab_mode: stabilize-all
regen_initramfs: true

# Dual deploy: libvirt + KubeVirt
emit_domain_xml: true           # Generate libvirt XML
deploy_k8s: true                # Deploy to KubeVirt
k8s_namespace: default          # Target namespace

vm_name: photon-kubevirt
memory: 2048
vcpus: 2
```

```
machine: q35
```

Single command, dual deploy: `sudo KUBECONFIG=/etc/rancher/k3s/k3s.yaml h2kvmctl --config photon-kubevirt.yaml`
Produces both libvirt domain XML and a running KubeVirt VirtualMachine on K3s.

Step 3: Migration Pipeline

STEP 3a — Flatten + Offline Fixes

```
14:20:03 Flatten snapshot chain
14:20:03 Flattening progress: 0.3%
14:20:05 Flattening progress: 5.6%
14:20:08 Flattening progress: 10.8%
14:20:08 Flattened: working-flattened.qcow2

14:20:08 Image validated: qcow2 (virtual: 8.00 GiB, actual: 0.97 GiB)
14:20:09 Connected to /dev/nbd0
14:20:09 Detected: VMware Photon OS 5.0 (photon, kernel 6.1.10-11.ph5)
14:20:09 Boot mode: UEFI
14:20:12 initramfs rebuild: dracut --add-drivers virtio_blk virtio_scsi virtio_net nvme
14:20:25 Systemd boot integration applied: 3 features
```

STEP 3b — Convert + Dual Deploy

```
14:20:25 Convert to QCOW2
14:20:25 Converting: working-flattened.qcow2 -> photon.qcow2
14:20:27 Validated: photon.qcow2

14:20:27 emit_domain_xml: guest=linux uefi=True name=photon-kubevirt
14:20:27 Domain XML: out-kubevirt-demo/libvirt/photon-kubevirt.xml

14:20:29 Creating PVC: photon-disk
14:20:29 PVC already exists: photon-disk
14:20:29 Uploading image to PVC
14:20:33 Copying photon.qcow2 to PVC (this may take a while)...
14:20:44 Image uploaded to PVC
14:20:44 Kubernetes deployment complete
14:20:44    PVC: photon-disk
```

Libvirt Output

- Domain XML: `photon-kubevirt.xml`
- Machine: Q35 with PCIe
- Firmware: UEFI (OVMF)

KubeVirt Output

- VirtualMachine: `photon` in `default` ns
- PVC: `photon-disk` (QCOW2 uploaded via CDI)
- Disk bus: VirtIO

- Disk: VirtIO (vda)
- Network: VirtIO
- Ready for: `virsh define`

- Network: masquerade (pod network)
- Labels: `hyper2kvm.io/migrated: "true"`
- Ready for: `kubectl patch vm ... running:true`

Step 4: KubeVirt VM Running

STEP 4a — VirtualMachine Created

```
$ kubectl get vm
NAME     AGE   STATUS    READY
photon   69s   Stopped   False

$ kubectl get vm photon -o yaml
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  labels:
    app: photon
    hyper2kvm.io/migrated: "true"
  name: photon
  namespace: default
spec:
  template:
    spec:
      architecture: amd64
      domain:
        devices:
          disks:
          - disk:
              bus: virtio
            name: rootdisk
          interfaces:
          - masquerade: {}
            name: default
        machine:
          type: q35
        resources:
          requests:
            cpu: "2"
            memory: 2Gi
      networks:
      - name: default
        pod: {}
      volumes:
      - name: rootdisk
        persistentVolumeClaim:
          claimName: photon-disk
```

STEP 4b — Start VM & Verify

```
$ kubectl patch vm photon --type merge -p '{"spec":{"running":true}}'
virtualmachine.kubevirt.io/photon patched

$ kubectl get vmi -o wide
NAME      AGE    PHASE      IP            NODENAME                  READY
photon    15s    Running    10.42.0.25    ssahani-thinkpad...       True

$ kubectl get vmi photon -o jsonpath='{.status.phase}'
Running

$ kubectl get vmi photon -o jsonpath='{.status.interfaces[0].ipAddress}'
10.42.0.25
```

VM Status

- Phase: **Running**
- Ready: **True**
- IP: 10.42.0.25
- Node: ssahani-thinkpad...
- Architecture: amd64

Resources

- CPU: 2 cores
- Memory: 2 GiB
- Disk: VirtIO (PVC-backed)
- Network: masquerade
- Machine: Q35

Storage

- PVC: photon-disk
- Image: photon.qcow2 (1 GB)
- Uploaded via CDI
- StorageClass: local-path
- Source: Photon OS 5.0 VMDK

Pipeline Summary

#	Step	Action	Duration	Result
1	Sanity checks	Tools, disk space, permissions	<1s	**5/5 passed**
2	Flatten	VMDK → QCOW2	~5s	**working-flattened.qcow2 (8 GiB virtual)**
3	Offline fixes	fstab, initramfs, GRUB, VMware cleanup	~17s	**VirtIO drivers, UEFI boot, Photon 5.0 detected**
4	Convert	Working → final QCOW2	~2s	**photon.qcow2 (1017 MB)**
5	Libvirt XML	Generate domain XML	<1s	**photon-kubevirt.xml (Q35, VirtIO, OVMF)**
6	PVC create	Create PersistentVolumeClaim	<1s	**photon-disk PVC**
7	PVC upload	Upload QCOW2 via CDI	~15s	**Image uploaded to PVC**
8	VM create	Apply VirtualMachine manifest	<1s	**photon VM in default namespace**
9	VM start	kubectl patch running:true	~15s	**Running, IP 10.42.0.25, Ready: True**

Output Files

```
out-kubevirt-demo/
  photon.qcow2              1017 MB   Final converted image
  photon-kubevirt.xml        1.8 KB   KubeVirt VirtualMachine manifest
  libvirt/
    photon-kubevirt.xml      1.8 KB   Libvirt domain XML (Q35, VirtIO, OVMF)
  work/
    working-flattened.qcow2  1013 MB   Intermediate image
```

Cluster State After Migration

```
$ kubectl get vm,vmi,pvc
NAME                              AGE   STATUS    READY
virtualmachine.kubevirt.io/photon 2m    Running   True

NAME                                         AGE   PHASE     IP           READY
virtualmachineinstance.kubevirt.io/photon     90s   Running   10.42.0.25   True

NAME                          STATUS   VOLUME        CAPACITY   STORAGECLASS
```

```
persistentvolumeclaim/photon   Bound    pvc-87dcd...   10Gi       local-path
```

VMware to KubeVirt — Single Command

Photon OS 5.0 VMDK (996 MB) → QCOW2 + offline fixes → PVC upload via CDI → KubeVirt VM running on K3s

VirtIO disk, Q35 machine, masquerade networking, IP assigned, Ready: True

Total time: ~44 seconds. One YAML. Zero kubectl commands needed.